EX. NO: 9

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR

AIM

To implement subnetting in Cisco Packet Tracer and configure a network using subnetted IP addresses.

SUBNETTING

The network 172.16.0.0/24 is subnetted using a /27 mask, giving 8 subnets with 30 usable hosts each.

Device	IP Address
Router Fa0/0	172.16.0.1
Router Fa0/1	172.16.0.33
PC1	172.16.0.2
PC2	172.16.0.3
PC3	172.16.0.34
PC4	172.16.0.35

PROCEDURE

1. Place a router, 2 switches and 4 PCs in Packet Tracer.
2. Connect PC1, PC2 to Switch1 and Switch1 to Router Fa0/0.
3. Connect PC3, PC4 to Switch2 and Switch2 to Router Fa0/1.
4. Configure the router interfaces and assign IP addresses to all devices as per the table.
5. Enable both interfaces using no shutdown.
6. Ping from PC1 to PC3 to check connectivity across subnets.

OUTPUT

Ping from PC1 to PC3 was successful, showing that devices in different subnets could communicate through the router.

RESULT

Subnetting was implemented successfully in Cisco Packet Tracer and connectivity between subnets was verified.